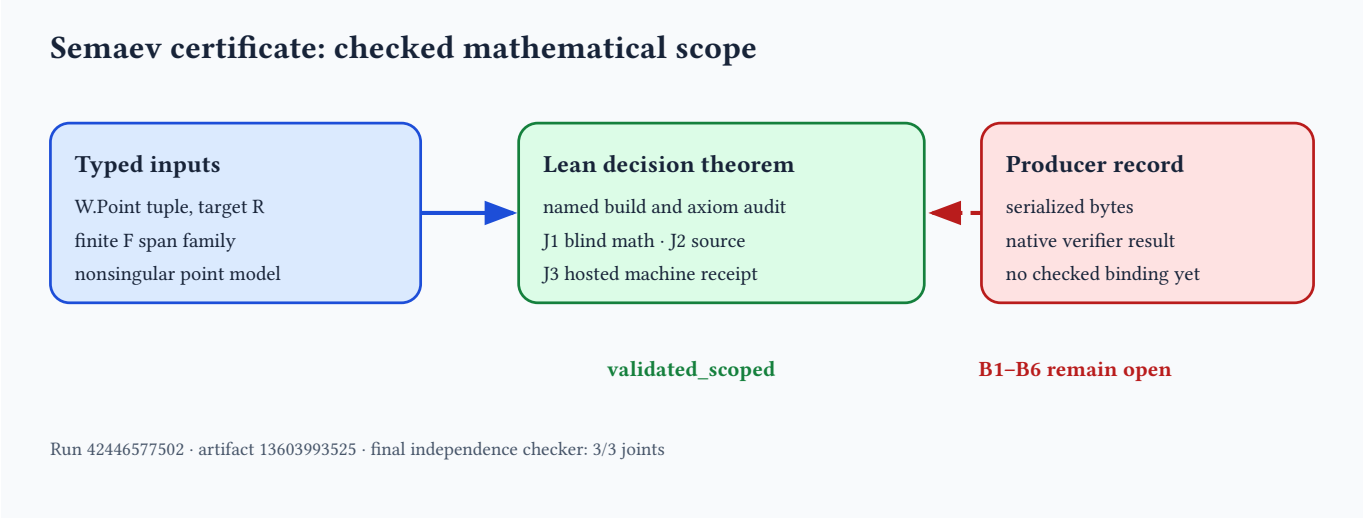
Semaev decomposition certificate

Prospective independent review · 9 November 2026

The typed finite-decomposition decision theorem passed a named hosted Lean build and axiom audit. A blind mathematical reviewer derived its predicate and discriminating F5 controls; an independent source reviewer matched the scoped theorem. The separate interpretation from producer record bytes to the theorem's arguments remains open.



Boundary	Evidence	Review outcome
Typed nonsingular `W.Point` tuple over supplied finite span	Named build, axiom audit; blind math and source checks	**validated_scoped** for that predicate
Raw `K × K` lemma under `W.IsElliptic`	Module compiled; no separate named raw audit	**inconclusive** pending target audit
Serialized producer certificate / native verifier	No checked interpretation or execution bridge	**needs_revision**; B1–B6 open

Evidence and controls

Plan commit `8a7434f59f3795d2f803ddaafc665e917aaf44c0` was published before dispatch. Hosted run **42446577502**, formal job **128626693105**, and artifact **13603993525** bind the named declaration `CryptoResearch.Semaev.decidable_validDecomposition`; artifact ZIP SHA-256 is `181433aeee8e030b0d33787e4d72e95d69ca9e59212abd6de864a48257cf3897`. The audit reports 1,313 theorems with standard axioms only. J1 checked empty tuple/span, off-curve input and wrong target. J3′s wrong-name and changed-hash controls failed as expected; a self-consistent synthetic receipt passed the consistency checker, so external run custody matters.

Independence and next construction

Three reviewers sealed separate reports. Their first J1/J2 YAML attestations had format defects; original bytes and seals are retained, and reviewer-issued additive metadata corrections leave the reports and verdicts unchanged. The final independence checker passes **3/3** owned joints. Next, freeze the wire-to-Lean record contract, bind one immutable sample to field, curve, span, tuple and target, then execute the instantiated decision with authenticated evidence. The raw-coordinate lemma needs a separate named audit.

Source: `technical-review.md`; plan: `review-plan.yaml`; final checker: `independence-check.txt`. No measured performance or official scientific status changed.